Project 4: Fine-Tune Your Own Chatbot

Welcome to Project 4. You'll build and fine-tune a simple chatbot with a clear domain or personality, then make it usable for real interactions.

WHAT YOU'LL BUILD

A domain-specific or personality-driven chatbot fine-tuned from a pretrained language model.

Get creative:

- Pick a clear niche (university guide, movie recommender, OMC assistant, funny persona, etc.)
- Define a strong personality or role (formal, friendly, sarcastic, mentor-style)
- Optional: add a simple web or GUI interface so others can chat with it

RESOURCES

Key concepts to research :

- Pretrained LLMs (GPT-style, LLaMA-based, Mistral, etc.)
- Fine-tuning vs prompting (instruction tuning)
- Dataset creation for dialogue (input → response pairs)
- Evaluation of chatbot responses

Optional learning resources :

- Hugging Face Course — Fine-tuning LLMs
- Stanford CS224N (language models sections)
- OpenAI / Hugging Face fine-tuning docs

2-WEEK SPRINT

Week 1: Design + Data + Fine-Tuning

- Define chatbot goal, domain, and personality
- Collect or generate a small dataset (50–200 examples)
- Format data correctly (JSON, instruction format, etc.)
- Fine-tune a pretrained model (lightweight setup)

- Mid-week check-in: **Wednesday** (ask questions, report progress)

Week 2: Interaction + Polish

- Test and refine chatbot behavior
- Improve responses (consistency, tone, usefulness)
- Add a simple interface (CLI, Streamlit, or web UI)
- Document limitations and future improvements

DELIVERABLES

- **Deadline:** January 1st at 23:59
- **Demo:** TBD

During the demo, you must:

- Present your working fine-tuned chatbot
- Explain your choice of domain and personality
- Describe the dataset creation process (source, size, format)
- Explain fine-tuning vs prompting and what you applied
- Demonstrate live interactions with the chatbot
- Discuss limitations, failure cases, and possible improvements

Don’t forget to have fun!

OMC AI Section Leadership: Faycal